

Semantic Routing in NDN

Phase 2: Architectural Deep Dive & Lit Review

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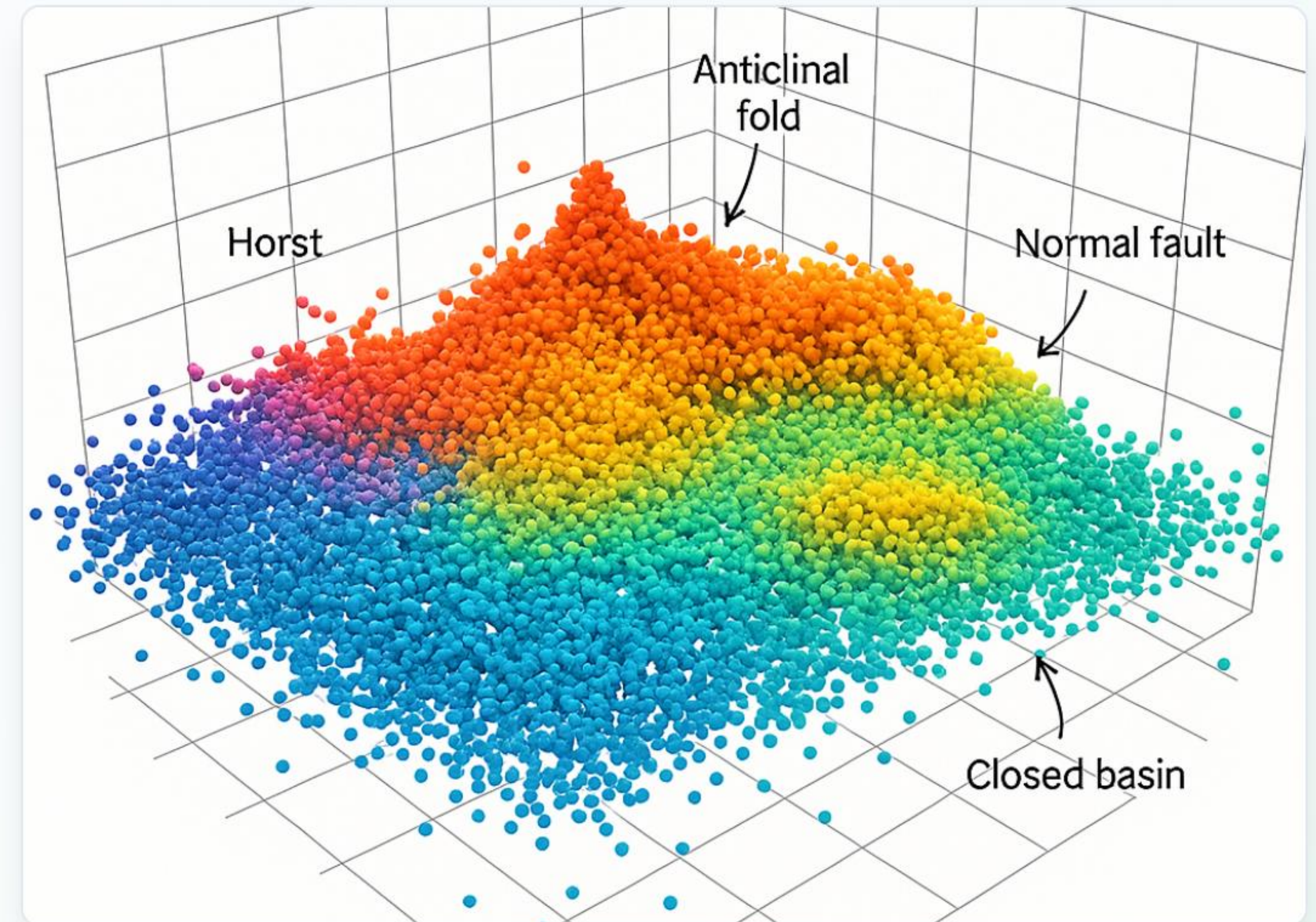
Phase 1 Recap: Concept-Based Forwarding

The Bottleneck

Standard NDN relies strictly on **Exact String Matching**. Minor naming differences (e.g., */dist-sys* vs */distributed-systems*) cause logical Cache Misses and packet drops.

Our Core Idea

An edge intelligence layer utilizing **Language Model Embeddings** (MiniLM) to map interests into a vector space, enabling mathematical similarity-based routing.



Isolation: Structure in Router Tables



FIB-Only Policy

Vector Similarity is restricted **only to the FIB**. Its purpose is to discover the best conceptual path when exact matching fails.



CS & PIT Stability

Content Store and PIT remain **strictly Exact-Match**. Applying semantic matching here risks delivering incorrect data to users.



Data Integrity

Ensures that Data Packets always follow the precise PIT chain back to the requester, preventing semantic-based logic errors.

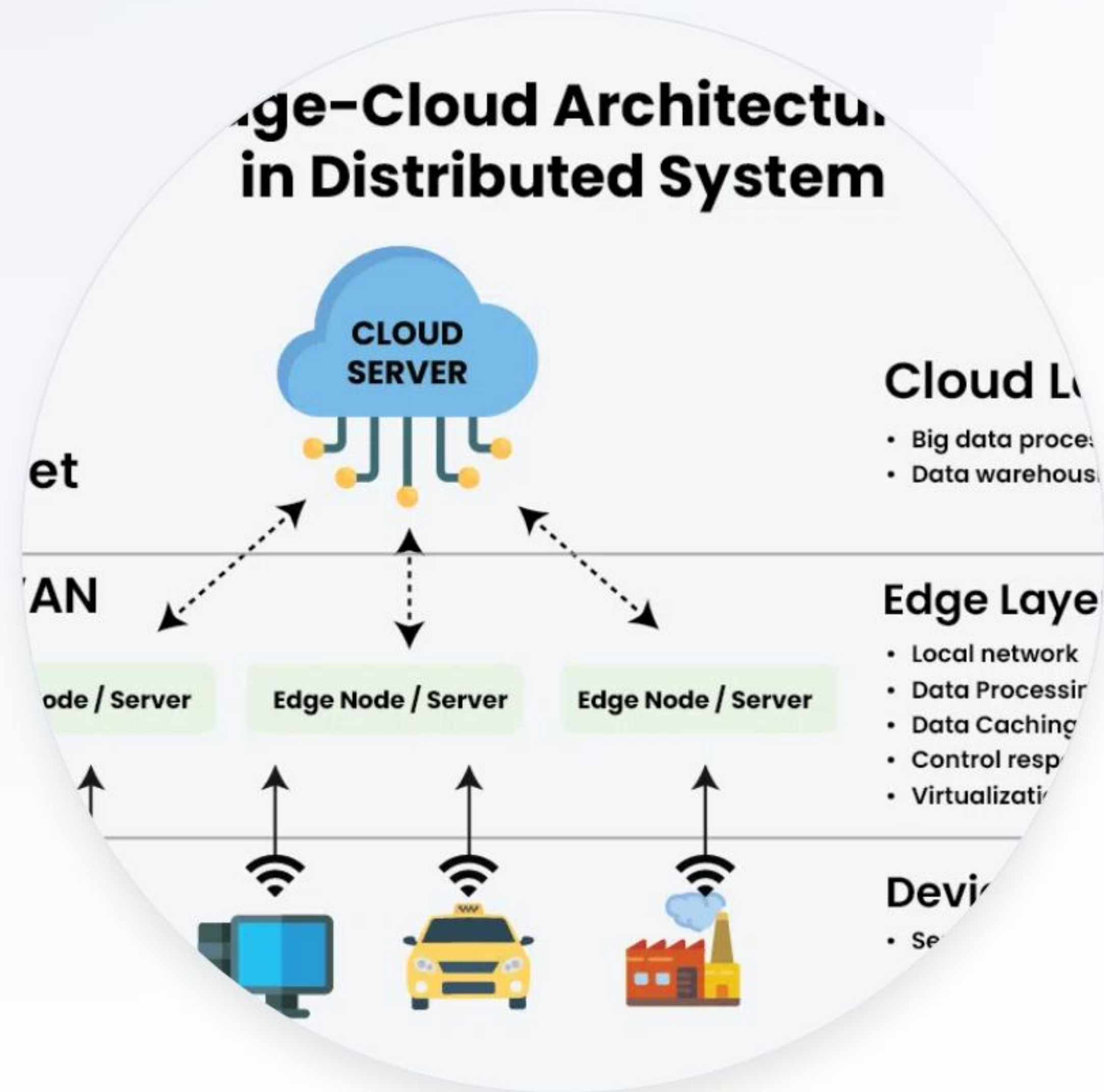
Latency: Line-Speed Optimization

Dual-Path Architecture

- **Fast Path (Nano-scale):** Standard hardware (TCAM) handles known traffic. AI is entirely bypassed for exact matches.
- **Slow Path (Milli-scale):** Only the first packet failure triggers FAISS discovery (approx. 2ms).
- **Convergence:** Discovery results are cached in the *Embedding Store*, allowing all subsequent packets to follow the Fast Path.

 High-performance data center server rack

Edge Acceleration & Quantization



Pruning Computational Cost

Edge Tagging: Vector calculations are offloaded to the first-hop edge router. Core routers simply process short *Semantic Tags* at nanosecond speeds.

Quantization: Converting heavy Float32 vectors to INT8 for direct execution on NPU chips, reducing inference time and power consumption by orders of magnitude.

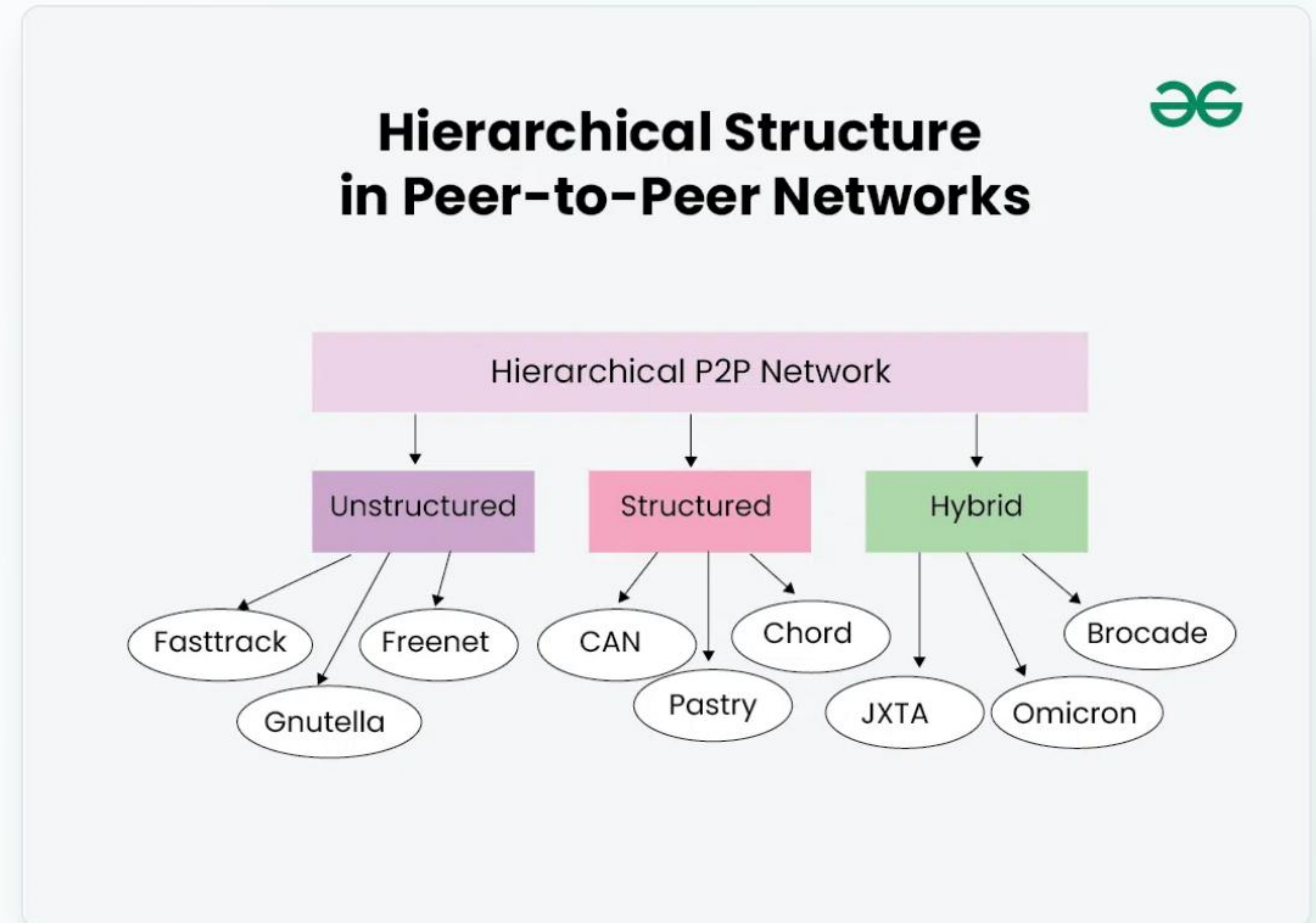
Article 1: SAF Strategy (IEEE IoT, 2026)

Core Methodology	Evaluation Metrics	Baseline Benchmark
Semantic-Aware Forwarding: Sentence Transformers (MiniLM) at Edge Routers + Cosine Similarity ($Th \geq 0.7$).	Service Discovery Time, pure Processing Time, and RAM overhead of AI models.	Vanilla NDN vs SAF: NDN failed completely on variants, while SAF reached 100% routing success.
Embedding Store (ES): LRU-based cache for vectors to prevent redundant AI computation.	Forwarding Effectiveness and Queue Latency under high request rates.	Raw SAF vs SAF+ES: ES resolved the router bottleneck, drastically reducing discovery delay.

Article 2: INF-NDN (IEEE Access, 2024)

Supernode-Based Architecture

- **Hierarchical Pruning:** Principal, Super, and Ordinary nodes form the backbone.
- **NLP Optimization:** Stemming, Stop-word removal, and abbreviation reduction.
- **LDA Tagging:** Latent Dirichlet Allocation assigns contextual tags (e.g.,) to nodes and packets.



Article 2: Efficiency vs. SICN

40.05%

Name Length Reduction

Key Performance Gains

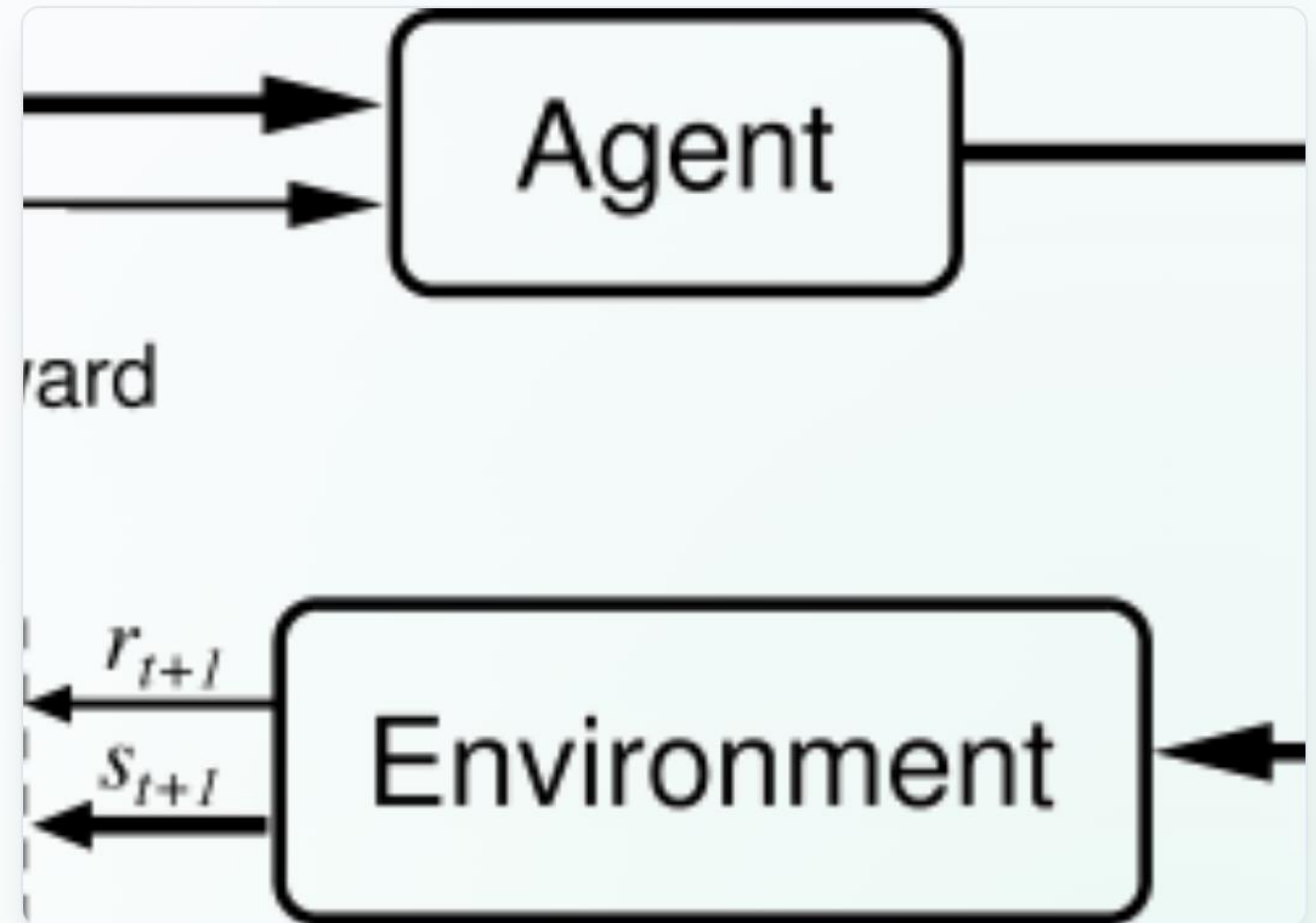
Memory Optimization: NLP cleanup compressed names, directly reducing FIB table memory footprints by 40% compared to SICN.

Latency: Interest Retrieval Time (IRT) dropped by **78-90%** as the system avoids broadcasts through direct Supernode tagging.

Article 3: SEF Strategy (CMC, 2024)

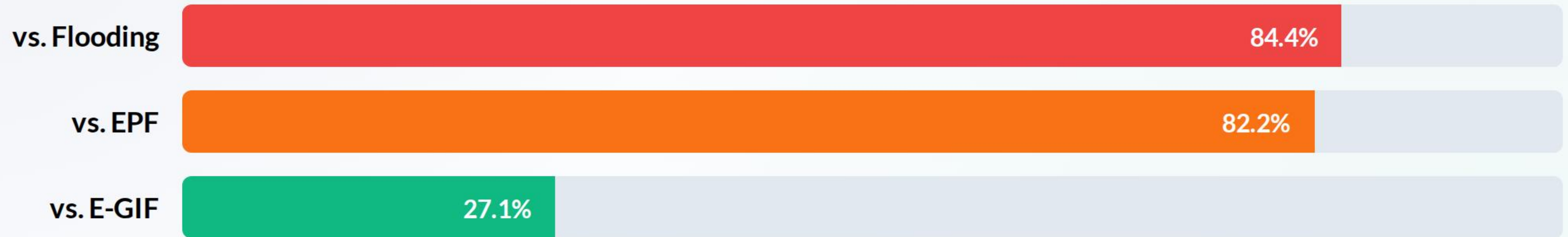
Energy-Aware Q-Learning

- **Dynamic Pathing:** Uses Q-Learning to select the next-hop based on Distance and Neighbor Energy.
- **FIB Removal:** Replaces heavy FIBs with lightweight *Neighbors* and *Producers* tables for IoT.
- **Header Pruning:** Adds *Next-Hop-ID* to explicitly eliminate duplicate transmissions (collisions).



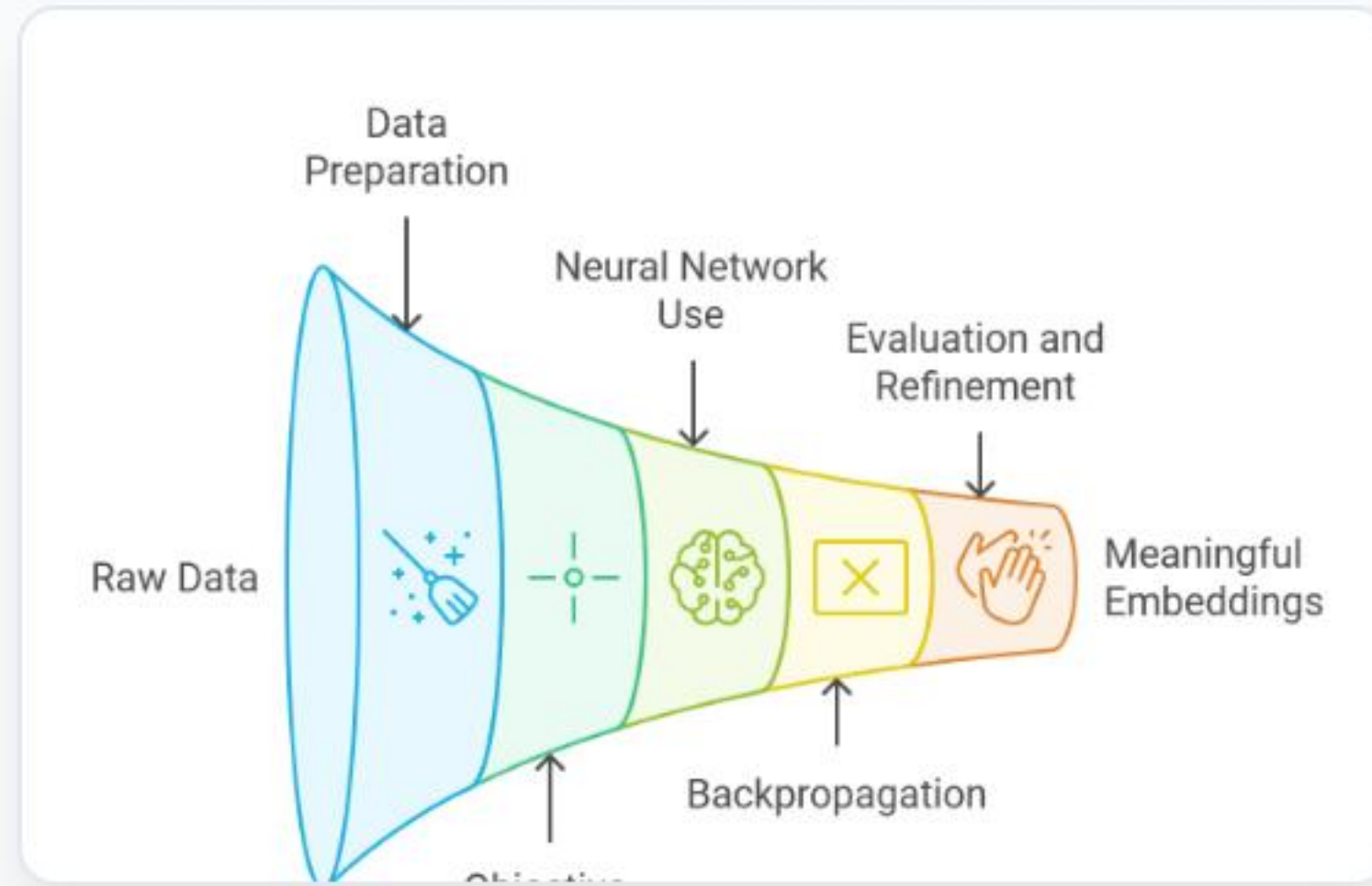
Article 3: Energy Savings Benchmarks

Energy Consumption Reduction (%)



SEF achieved a Packet Delivery Ratio (PDR) 0.69 higher than baselines in dense 4000+ node topologies, extending lifetime by 65%.

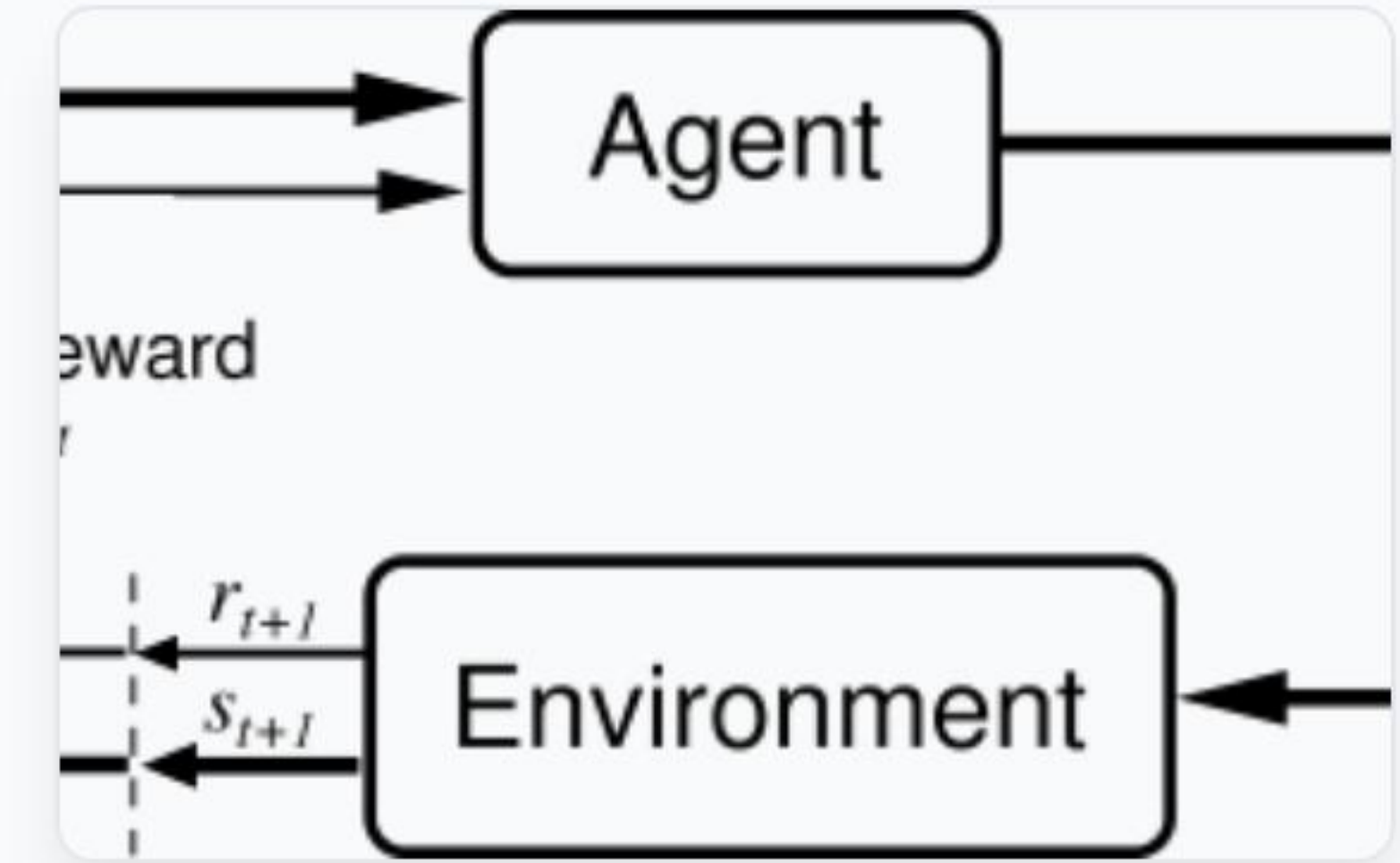
Building Our Unified Framework



Validation (SAF+ES): Proves our Sentence Transformer approach is viable if vector caching is strictly maintained.

 Abstract digital network connectivity

Scalability (INF-NDN): Guides our LDA-based tagging to ensure Core routers remain at Line-Speed.



Strategy (SEF): Inspires our Neighbor Tables to mitigate broadcast storms and conserve energy.

Final Vision: Line-Speed Convergence

99%

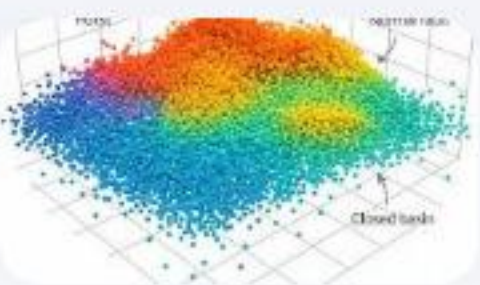
Nano-scale Convergence

The Final Benchmark (ns-3)

We aim to model the Fast/Slow Path layering. As the Embedding Store "warms up" (Cache Hit Ratio > 95%), total network latency will converge back to nanoseconds.

Baseline Targets: We will compare our *Semantic NDN* against *Vanilla NDN* and *SICN* regarding IRT, PDR, and energy efficiency.

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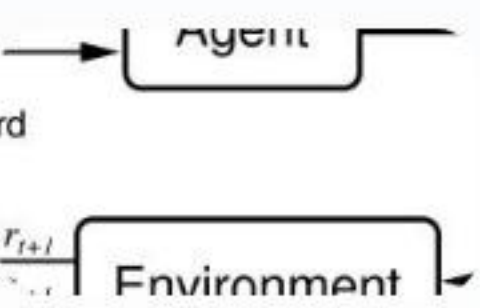
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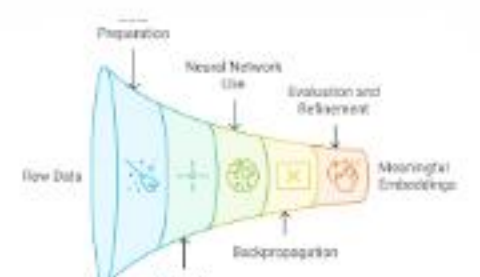
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